Serverless Image Processing Pipeline

AWS Architecture & Project Documentation

This document outlines the architecture, component justifications, and execution flow of a highly resilient, event-driven serverless image processing pipeline built natively on AWS. This documentation serves as the comprehensive technical guide for project submission.

1. Architecture Execution Flow (Step-by-Step)

1. **Authentication & Upload Request:** The end-user makes an API call to **Amazon API Gateway** to request image upload capabilities.
2. **Pre-signed URL Generation:** API Gateway triggers a lightweight **AWS Lambda** function that generates an S3 Pre-signed URL. This URL is returned directly to the user, authorizing a secure, temporary upload window.
3. **Direct Upload:** The user uploads the high-resolution image directly to the **S3 Source Bucket** using the pre-signed URL. This bypasses API Gateway's 10MB payload limits.
4. **Event Generation:** The S3 Source Bucket triggers an S3 Event Notification upon object creation, instantly sending a message payload to an **Amazon SQS Queue**.
5. **Event Polling & Routing: Amazon EventBridge Pipes** continuously polls the SQS queue. Upon reading a message, it seamlessly triggers an execution of the **AWS Step Functions** workflow.
6. **Workflow Orchestration:** Step Functions orchestrates the core processing pipeline using single-responsibility compute:
    - **Lambda (Validation):** Checks image format, size, and integrity.
    - **Lambda (Resize):** Resizes the image into required dimensions (utilizing Lambda Layers for image binaries).
    - **Lambda (Watermark):** Applies a visual watermark to the resized image.
7. **Storage & Delivery:** The final processed image is stored securely in the **S3 Destination Bucket**. S3 Lifecycle Policies are applied here to automatically transition older objects to cheaper storage classes over time.
8. **Metadata & Notification:** The Step Functions workflow uses *Direct Service Integrations* to write the image metadata to **Amazon DynamoDB** and publish a job status (success/failure) to an **Amazon SNS Topic**, immediately alerting the Administrator.
9. **Content Delivery:** End-users view the processed images globally via **Amazon CloudFront**, which securely accesses the private S3 Destination Bucket via Origin Access Control (OAC).